

CHAPTER 2

ARTIFICIAL NEURAL NETWORKS: AN INTRODUCTION

DEFINITION OF NEURAL NETWORKS

According to the DARPA Neural Network Study (1988, AFCEA International Press, p. 60):

- ... a neural network is a system composed of many simple processing elements operating in parallel whose function is determined by network structure, connection strengths, and the processing performed at computing elements or nodes.

According to Haykin (1994), p. 2:

A neural network is a massively parallel distributed processor that has a natural propensity for storing experiential knowledge and making it available for use. It resembles the brain in two respects:

- Knowledge is acquired by the network through a learning process.
- Interneuron connection strengths known as synaptic weights are used to store the knowledge.

NEURAL NETWORKS

- Neural network was inspired by the design and functioning of human brain and components.

Definition:

- Information processing model that is inspired by the way biological nervous system (i.e) the brain, process information.
- ANN is composed of large number of highly interconnected processing elements(neurons) working in union to solve problems.
- It is configured for special application such as pattern recognition and data classification through a learning process.
- 85-90% accurate.

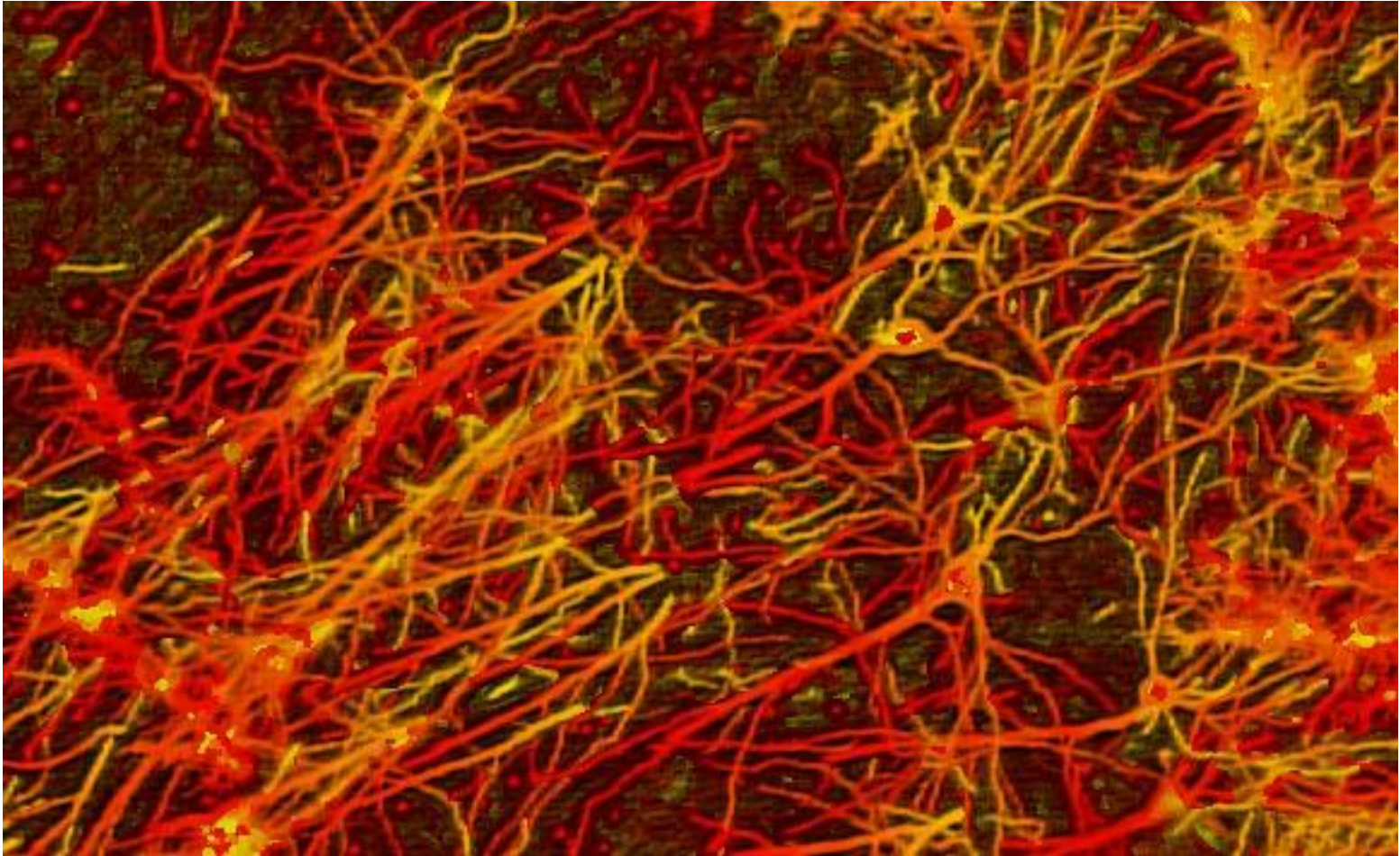
BRAIN COMPUTATION



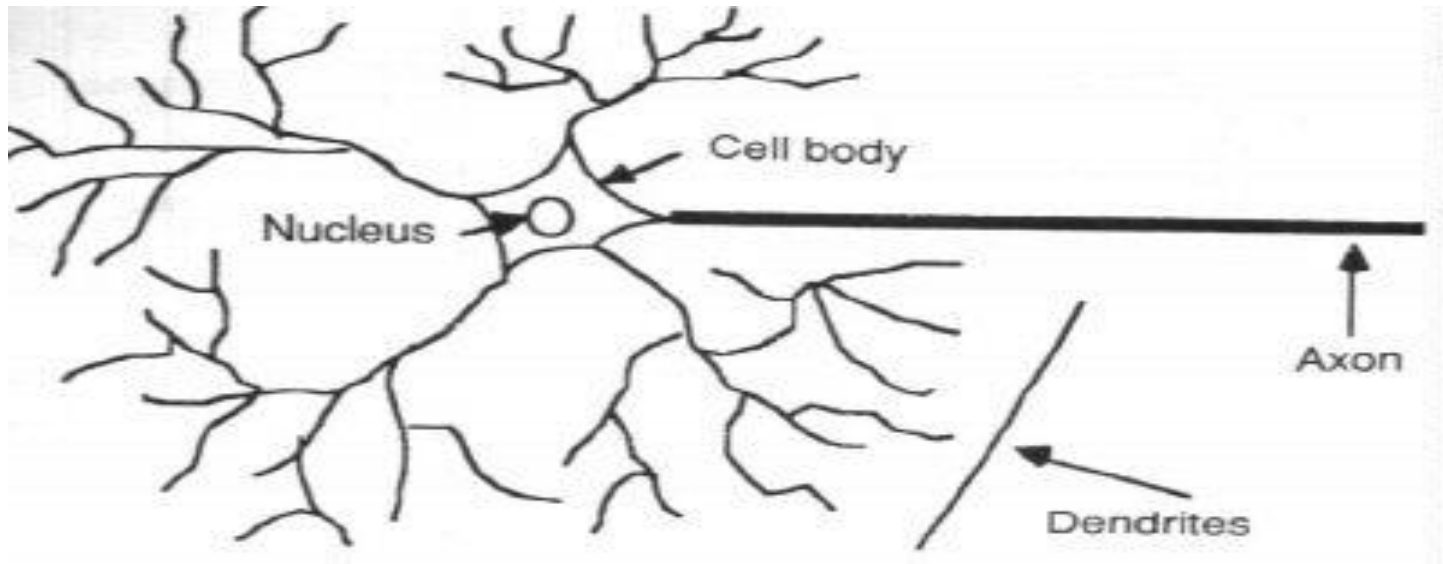
The **human brain** contains about 10 billion nerve cells, or neurons. On average, each neuron is connected to other neurons through approximately 10,000 synapses.

	processing elements	element size	energy use	processing speed	style of computation	fault tolerant	learns	intelligent, conscious
	10^{14} synapses	10^{-6} m	30 W	100 Hz	parallel, distributed	yes	yes	usually
	10^8 transistors	10^{-6} m	30 W (CPU)	10^9 Hz	serial, centralized	no	a little	not (yet)

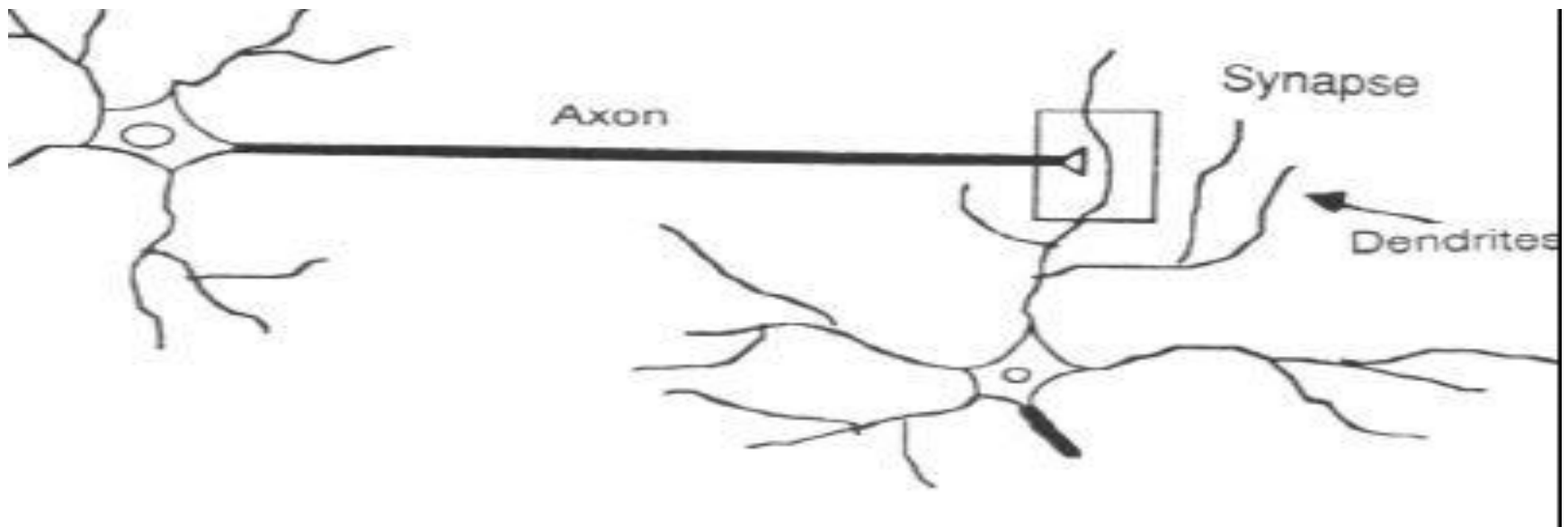
INTERCONNECTIONS IN BRAIN



BIOLOGICAL NEURAL NETWORK



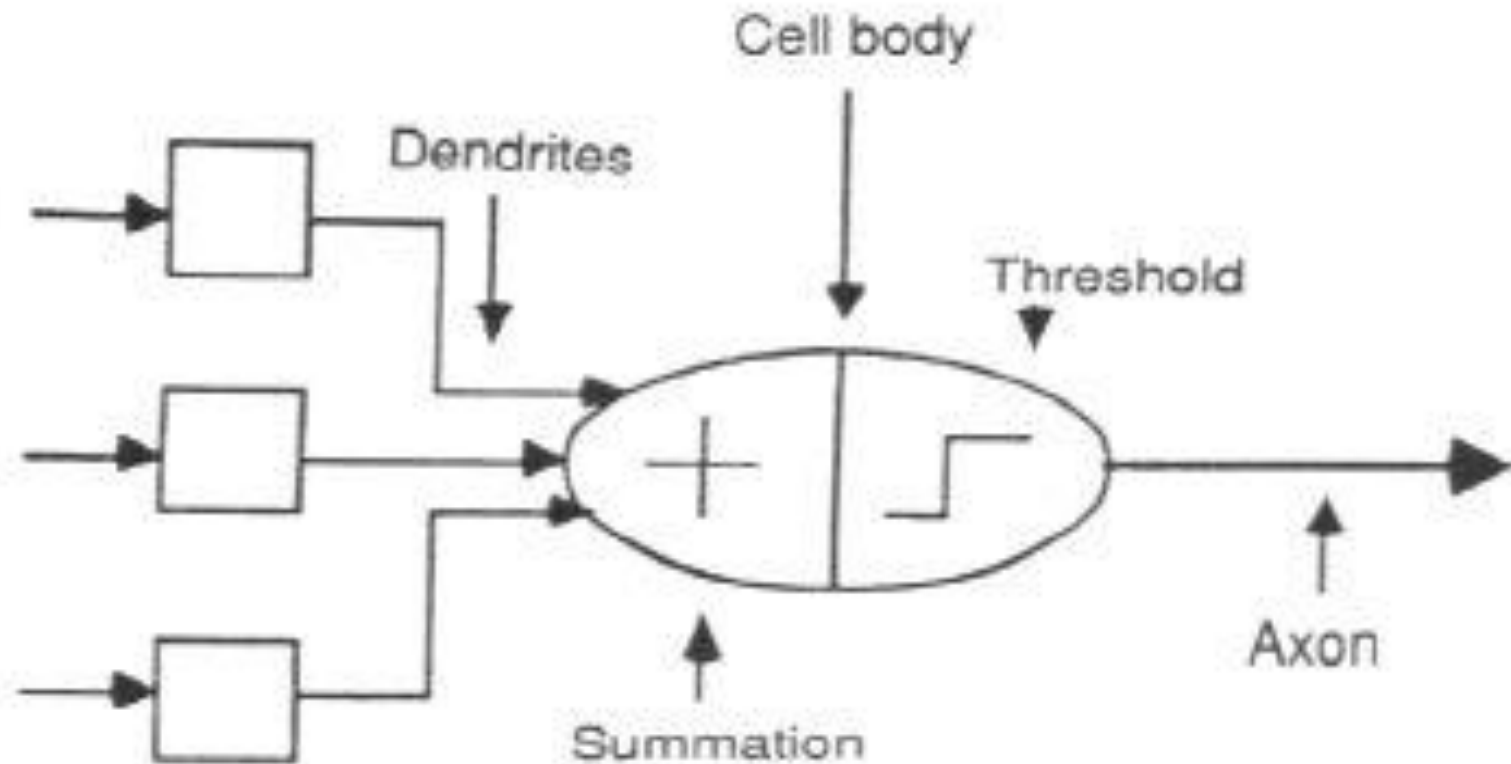
- Human brain consist of a huge number of neurons, appx 10^{11} , with numerous interconnections
- Consist of 3 main parts
 - Soma/cell body *where cell nucleus is located*
 - Dendrites *where nerve is connected to cell body*
 - Axon *which carries the impulse of neuron*



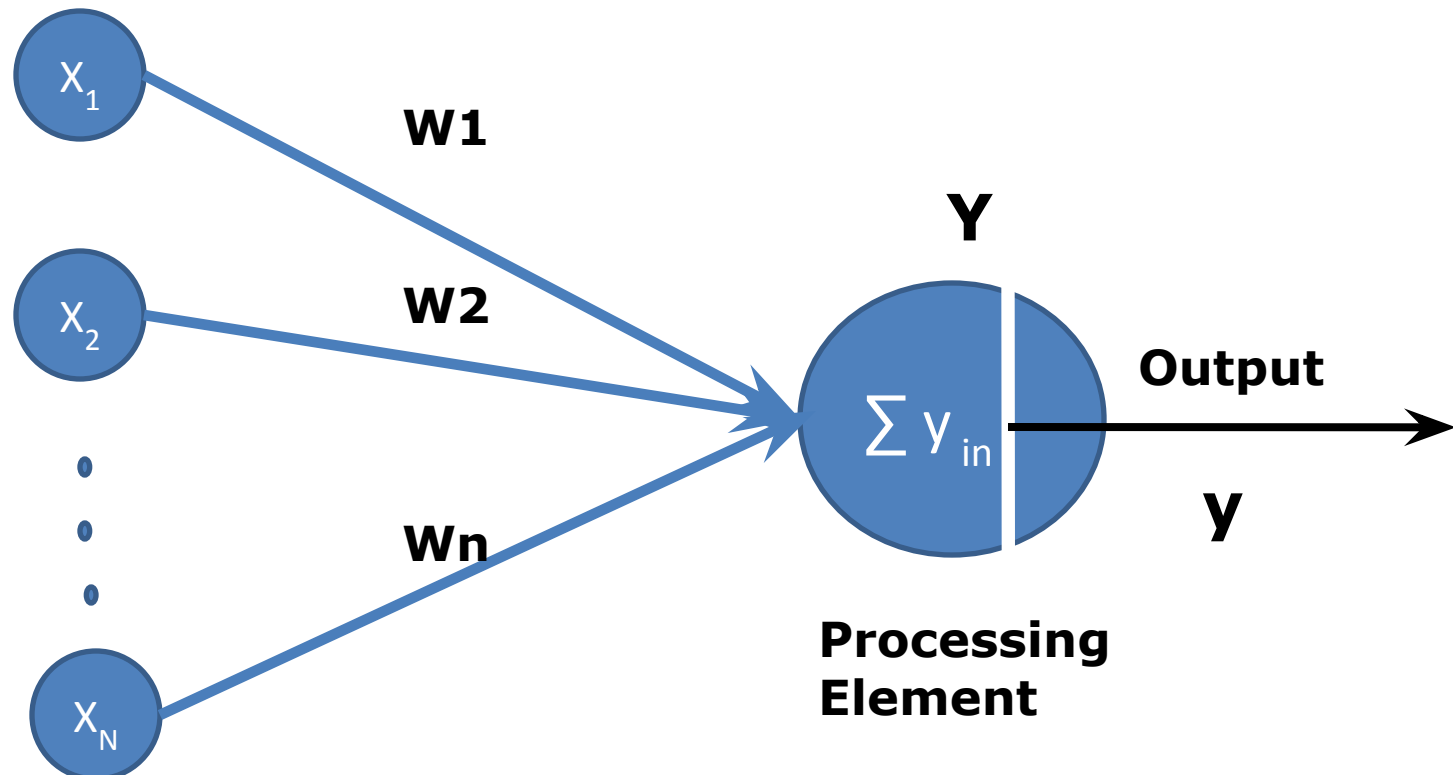
- ✓ In the human brain, a typical neuron collects signals from others through a host of fine structures called *dendrites*.
- ✓ The neuron sends out spikes of electrical activity through a long, thin strand known as an *axon*, which splits into thousands of branches.
- ✓ At the end of each branch, a structure called a *synapse* converts the activity from the axon into electrical effects that *inhibit* or *excite* activity in the connected neurons.

- Electric impulse is passed between synapse and dendrites.
- **Synapse**- Axon split into strands and strands terminates into small bulb like organs called as synapse.
- It is a chemical process which results in increase /decrease in the electric potential inside the body of the receiving cell.
- If the electric potential reaches a threshold value, receiving cell fires & pulse / action potential of fixed strength and duration is send out through the axon to synaptic junction of the cell.
- After that, cell has to wait for a period called **refractory period**.
- Synapses are said to be
 - **Inhibitory** if the parsing impulses prevent the firing of receiving cell
 - **Excitatory** If the parsing impulses cause the firing of cells

ASSOCIATION OF BIOLOGICAL NET WITH ARTIFICIAL NET



- ✓ Figure below shows the mathematical representation of above discussed chemical reaction taking place in an artificial neuron



- ✓ The net input is calculated as

$$Y_{in} = x_1w_1 + x_2w_2 + \dots + x_nw_n$$

$$\sum_{i=1}^n x_i w_i$$

- The weight represent the strength of the synapse connecting input and the output neuron
- A positive weight corresponds to the excitatory synapses
- A negative weight corresponds to the inhibitory synapses

ARTIFICIAL NEURAL NET

- Information-processing system.
- Neurons process the information.
- The signals are transmitted by means of connection links.
- The links possess an associated weight.
- The output signal is obtained by applying activations to the net input.

MOTIVATION FOR NEURAL NET

- ❑ Scientists are challenged to use machines more effectively for tasks currently solved by humans.
- ❑ Symbolic rules don't reflect processes actually used by humans.
- ❑ Traditional computing excels in many areas, but not in others.

The major areas being:

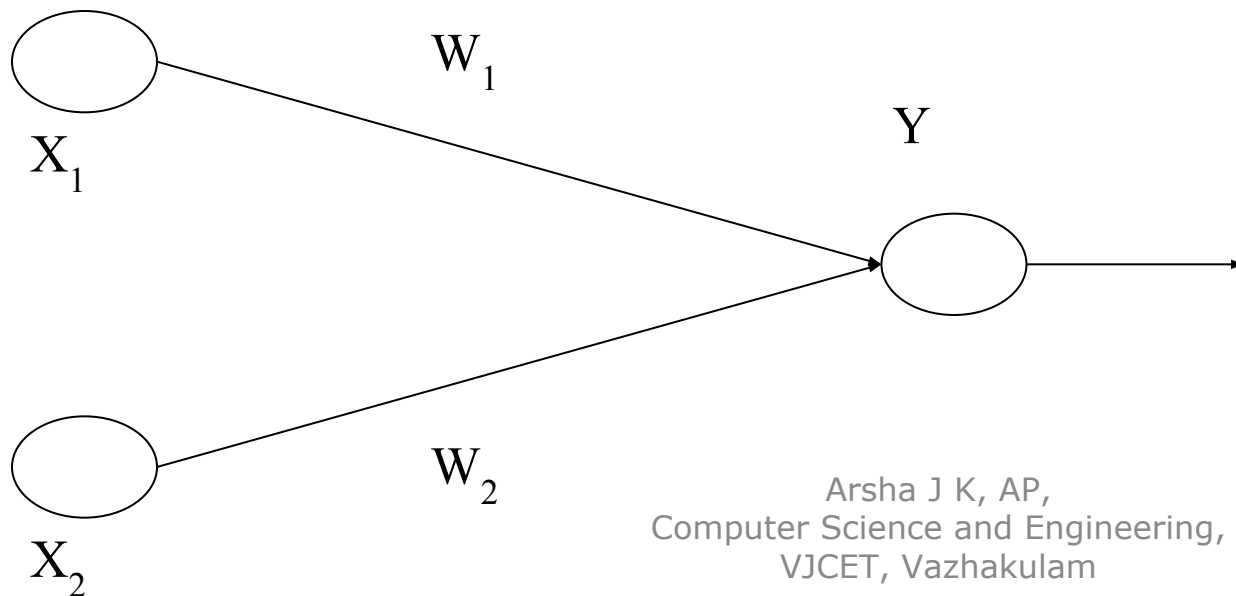
- Massive parallelism
- Distributed representation and computation
- Learning ability
- Generalization ability
- Adaptivity
- Inherent contextual information processing
- Fault tolerance
- Low energy consumption.

ARTIFICIAL NEURAL NETWORK : AN INTRODUCTION

- Resembles the characteristic of biological neural network.
- **Nodes** – interconnected processing elements (units or neurons)
- Neuron is connected to other by a **connection link**.
- Each connection link is associated with **weight** which has information about the input signal.
- ANN processing elements are called as **neurons or artificial neurons** , since they have the capability to model networks of original neurons as found in brain.
- Internal state of neuron is called **activation or activity level** of neuron, which is the function of the inputs the neurons receives.
- Neuron can send only one signal at a time.

BASIC OPERATION OF A NEURAL NET

- X_1 and X_2 – input neurons.
- Y - output neuron
- Weighted interconnection links- W_1 and W_2 .
- Net input calculation is :
- Output is : $Y_{in} = -x_1w_1 + x_2w_2$ $y = f(Y_{in})$
- Output= function (net input calculated)



- The function to be applied over the net input is called ***activation function***.
- Weight involved in ANN is equal to the slope of linear straight line ($y=mx$).



- To obtain y , the slope of m is directly multiplied with the input signal.
- This is linear equation when the slope and input is linearly varied
- This shows that the weight involved in ANN is equivalent to the slope of linear straight line

Terminology Relation Between Biological And Artificial Neuron

Biological Neuron	Artificial Neuron
Cell	Neuron
Dendrites	Weights or interconnections
Soma	Net input
Axon	Output

BRAIN VS COMPUTER

Term	Brain	Computer
Speed	Execution time is few milliseconds	Execution time is few nano seconds
Processing	Perform massive parallel operations simultaneously	Perform several parallel operations simultaneously. It is faster than the biological neuron
Size and complexity	Number of Neuron is 10^{11} and number of interconnections is 10^{15} . So complexity of brain is higher than computer	It depends on the chosen application and network designer.
Storage capacity	i) Information is stored in interconnections or in synapse strength. ii) New information is stored without destroying old one. iii) Sometimes fails to recollect information	i) Stored in continuous memory location. ii) Overloading may destroy older locations. iii) Can be easily retrieved
Tolerance	i) Fault tolerant ii) Store and retrieve information even if interconnections fail iii) Accept redundancies	i) No fault tolerance ii) Information corrupted if the network connections are disconnected. iii) No redundancies
Control mechanism	Depends on active chemicals and neuron connections are strong or weak	CPU Control mechanism is very simple

CHARACTERISTICS OF ANN:

- Neurally implemented mathematical model
- Large number of processing elements called neurons exists here.
- Interconnections with weighted linkage hold informative knowledge.
- Input signals arrive at processing elements through connections and connecting weights.
- Processing elements can learn, recall and generalize from the given data.
- Computational power is determined by the collective behavior of neurons.
 - *ANN is a connection models, parallel distributed processing models, self-organizing systems, neuro-computing systems and neuro morphic system.*

EVOLUTION OF NEURAL NETWORKS

Year	Neural network	Designer	Description
1943	McCulloch and Pitts neuron	McCulloch and Pitts	Arrangement of neurons is combination of logic gate. Unique feature is threshold
1949	Hebb network	Hebb	If two neurons are active, then their connection strengths should be increased.
1958,1959, 1962,1988, 1960	Perceptron Adaline	Frank Rosenblatt, Block, Minsky and Papert Widrow and Hoff	Weights are adjusted to reduce the difference between the net input to the output unit and the desired output
1972	Kohonen self-organizing feature map	Kohonen	Inputs are clustered to obtain a fired output neuron.
1982,1984, 1985,1986, 1987	Hopfield network	John Hopfield and Tank	Based on fixed weights. Can act as associative memory nets
1986	Back propagation network	Rumelhart, Hinton and Williams	i) Multilayered ii) Error propagated backward from output to the hidden units

Year	Neural network	Designer	Description
1988	Counter propagation network	Grossberg	Similar to kohonen network
1987-1990	Adaptive resonance Theory(ART)	Carpenter and Grossberg	Designed for binary and analog inputs.
1988	Radial basis function network	Broomhead and Lowe	Resemble back propagation network , but activation function used is Gaussian function
1988	Neo cognitron	Fukushima	For character recogniton.

BASIC MODELS OF ANN

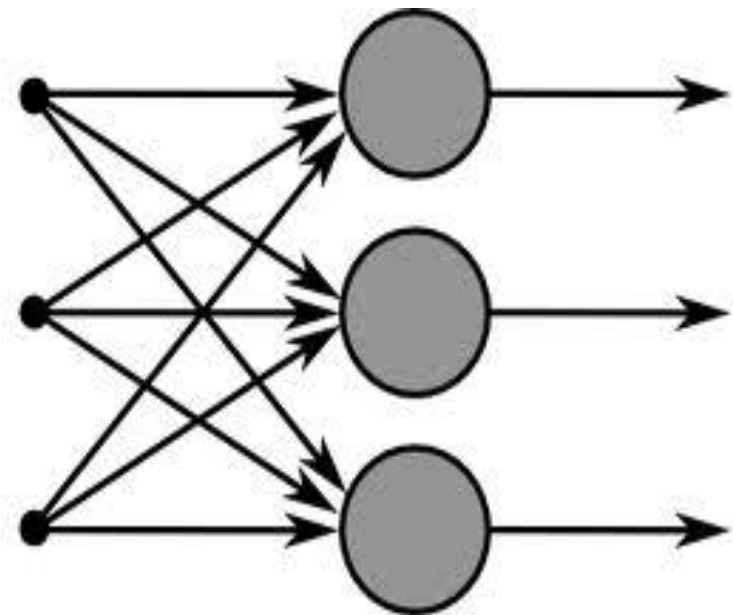
- ✓ Models are based on three entities
 - ❖ The model's *synaptic interconnections*.
 - ❖ The *training or learning rules* adopted for updating and adjusting the connection weights.
 - ❖ Their *activation functions*

CONNECTION

- The *arrangement of neurons to form layers* and the *connection pattern formed within and between layers* is called the ***network architecture***.
- Five types:
 - ***Single layer feed forward network***
 - ***Multilayer feed-forward network***
 - ***Single node with its own feedback***
 - ***Single-layer recurrent network***
 - ***Multilayer recurrent network***

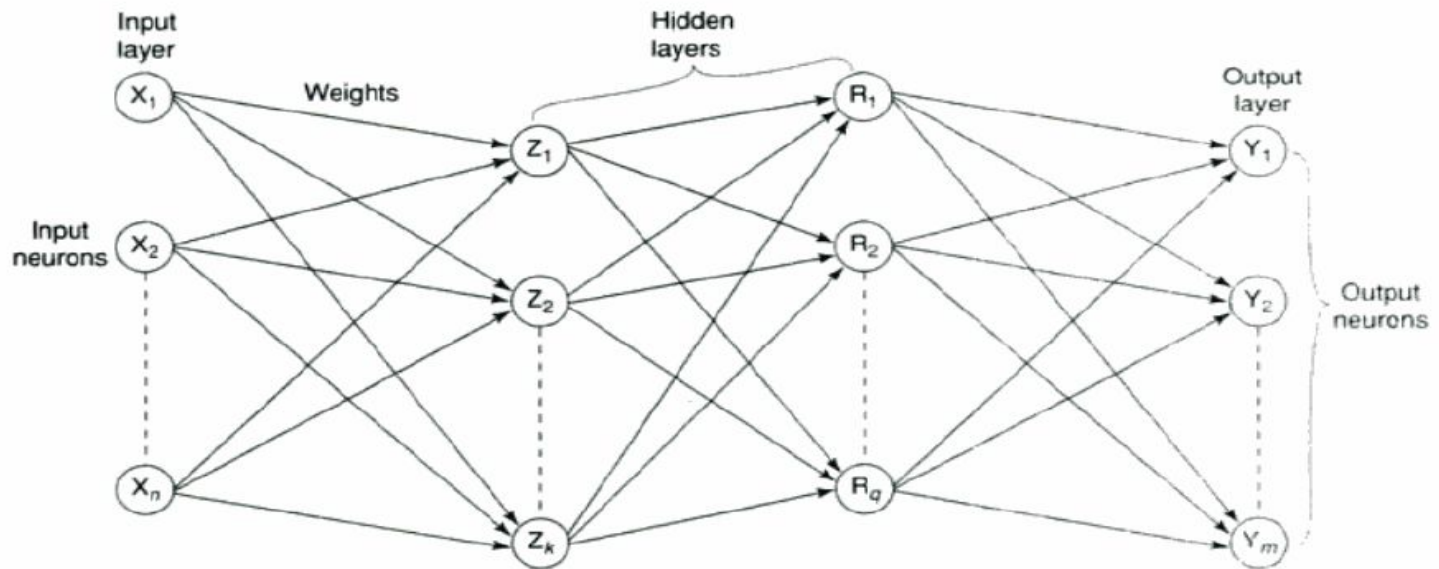
Single layer Feed-Forward Network

- Layer is formed by taking processing elements and combining it with other processing elements.
- Input and output are linked with each other
- Inputs are connected to the processing nodes with various weights, resulting in series of outputs one per node.



Multilayer feed-forward network

- Formed by the interconnection of several layers.
- Input layer receives input and buffers input signal.
- Output layer generated output.
- Layer between input and output is called *hidden layer*.
- Hidden layer is internal to the network.
- Zero to several hidden layers in a network.
- More the hidden layer, more is the complexity of network, but efficient output is produced.



FEED BACK NETWORK

- If no neuron in the output layer is an input to a node in the same layer / proceeding layer – **feed forward network**.
- If outputs are directed back as input to the processing elements in the same layer/proceeding layer – **feedback network**.
- If the output are directed back to the input of the same layer then it is **lateral feedback**.
- **Recurrent networks** are networks with feedback networks with closed loop.
- Fig 2.8 (A) –simple recurrent neural network having a single neuron with feedback to itself.
- Fig 2.9 – single layer network with feedback from output can be directed to processing element itself or to other processing element/both.

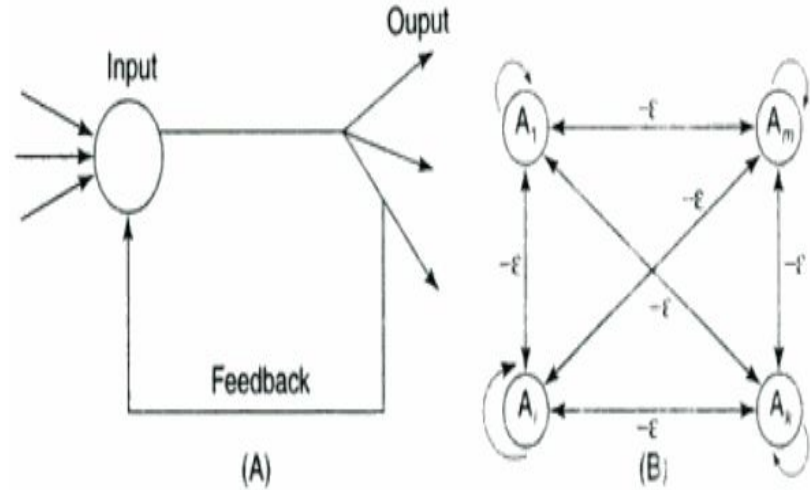


Figure 2-8 (A) Single node with own feedback. (B) Competitive nets.

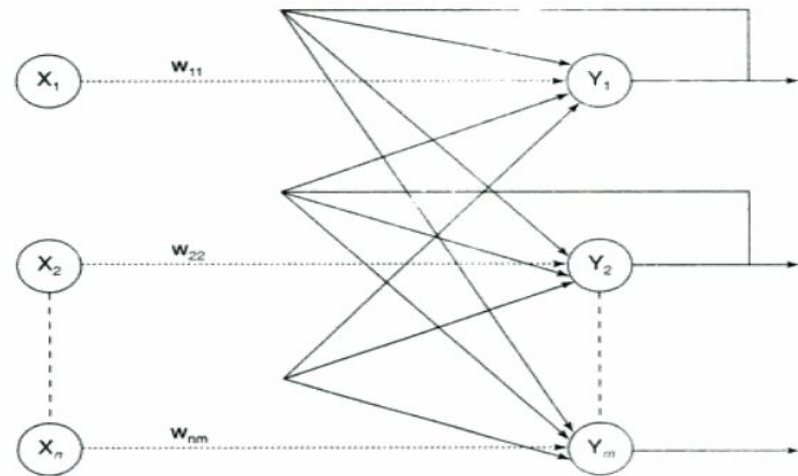
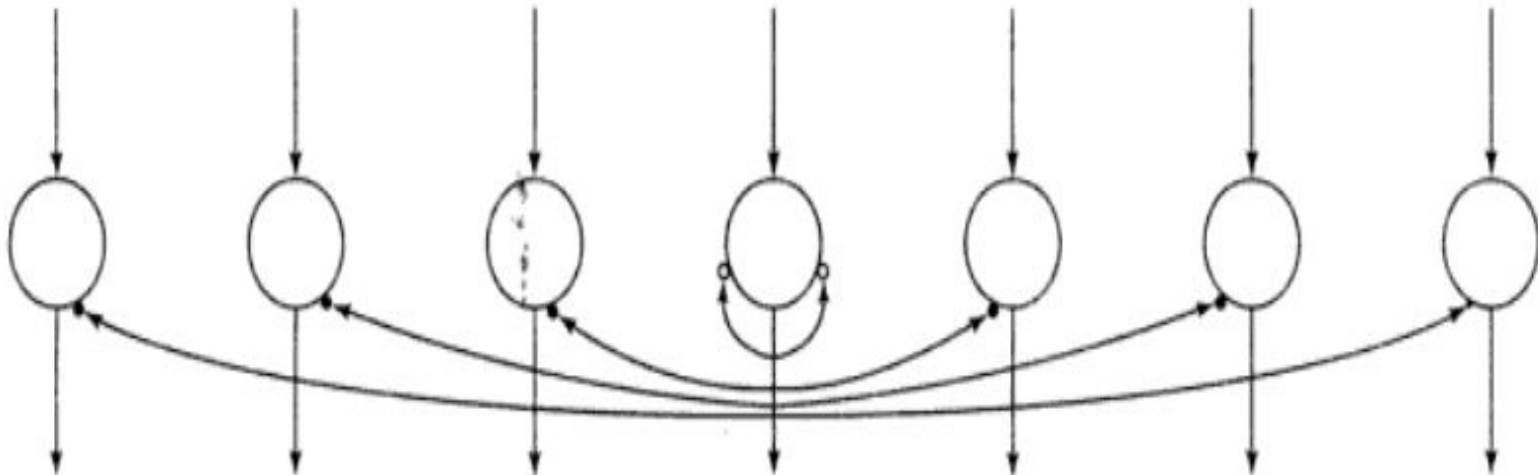
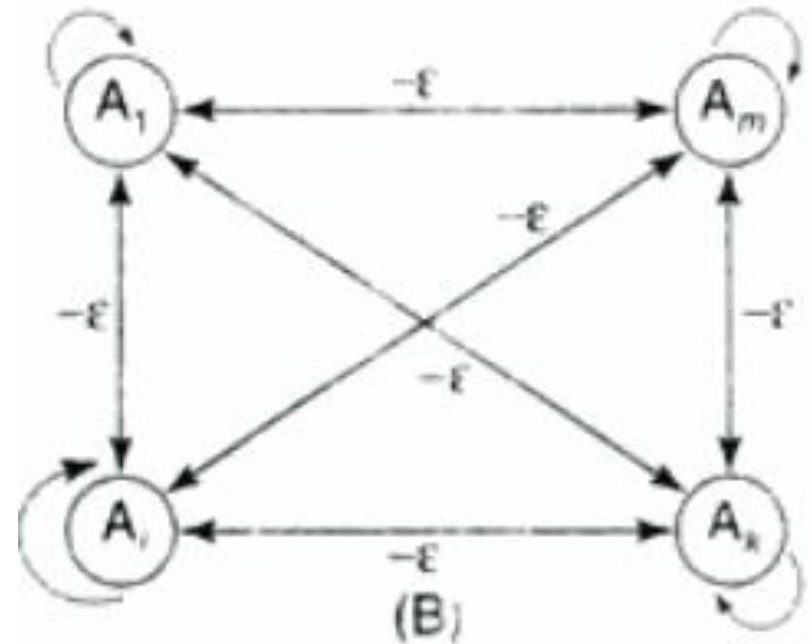
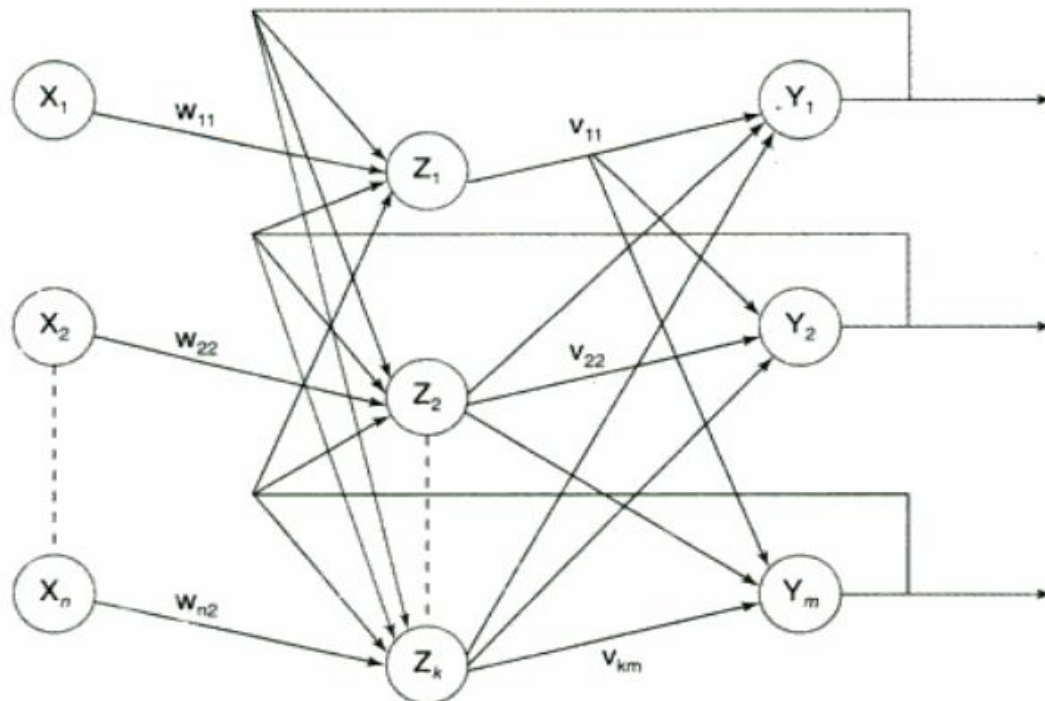


Figure 2-9 Single-layer recurrent network.

- **Maxnet** – competitive interconnections having fixed weights.
- **On-center-off-surround/lateral inhibition structure** – each processing neuron receives two different classes of inputs- “excitatory” input from nearby processing elements & “inhibitory” elements from more distantly located precessing elements. This type of interconnection is shown below



- Processing element output can be directed back to the nodes in the preceding layer, forming a *multilayer recurrent network*.
- Processing element output can be directed to processing element itself or to other processing element in the same layer.



LEARNING

- ✓ The main property of ANN
- ✓ NN adapt itself to a stimulus by making proper parameter adjustments resulting in the production of desired response
- ✓ Two broad kinds of learning in ANNs is :
 - ✓ *parameter learning* – updates connecting weights in a neural net.
 - ✓ *Structure learning* – focus on change in the network which include the number of processing elements and their connection type
- ✓ Apart from these, learning in ANN is classified into three categories as

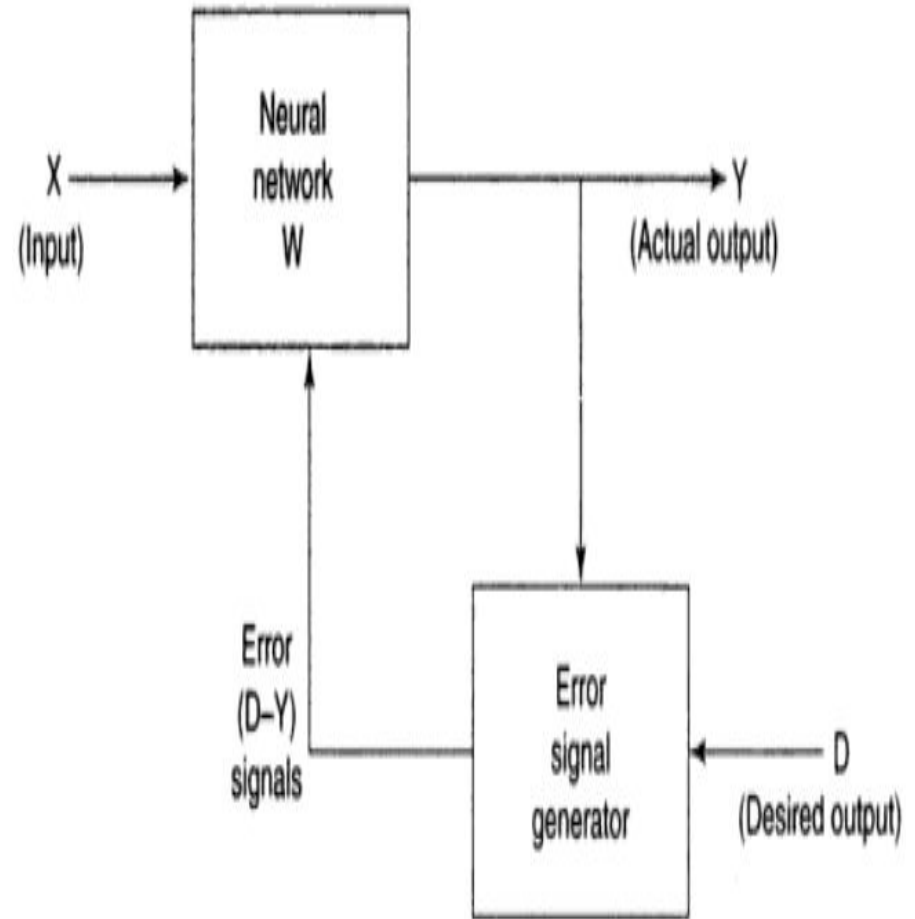
i) supervised learning

ii) unsupervised learning

Iii) reinforcement learning

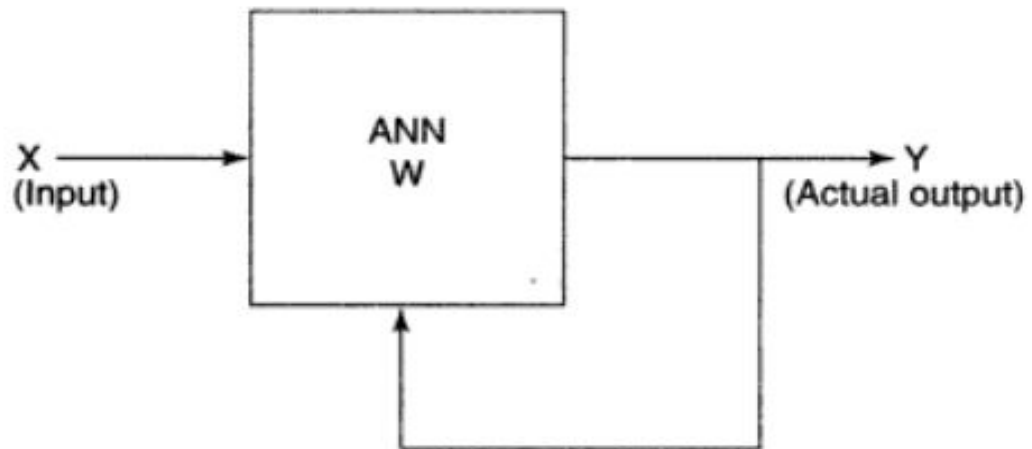
SUPERVISED LEARNING

- ✓ Learning with the help of a teacher.
- ✓ Example : learning process of a small child.
 - ✓ Child doesn't know read/write.
 - ✓ Their each & every action is supervised by a teacher
- ✓ In ANN, each input vector requires a corresponding target vector, which represents the desired output.
- ✓ The input vector along with target vector is called **training pair**.
- ✓ The input vector results in output vector.
- ✓ The actual output vector is compared with desired output vector.
- ✓ If there is a difference means an error signal is generated by the network.
- ✓ It is used for adjustment of weights until actual output matches desired output.



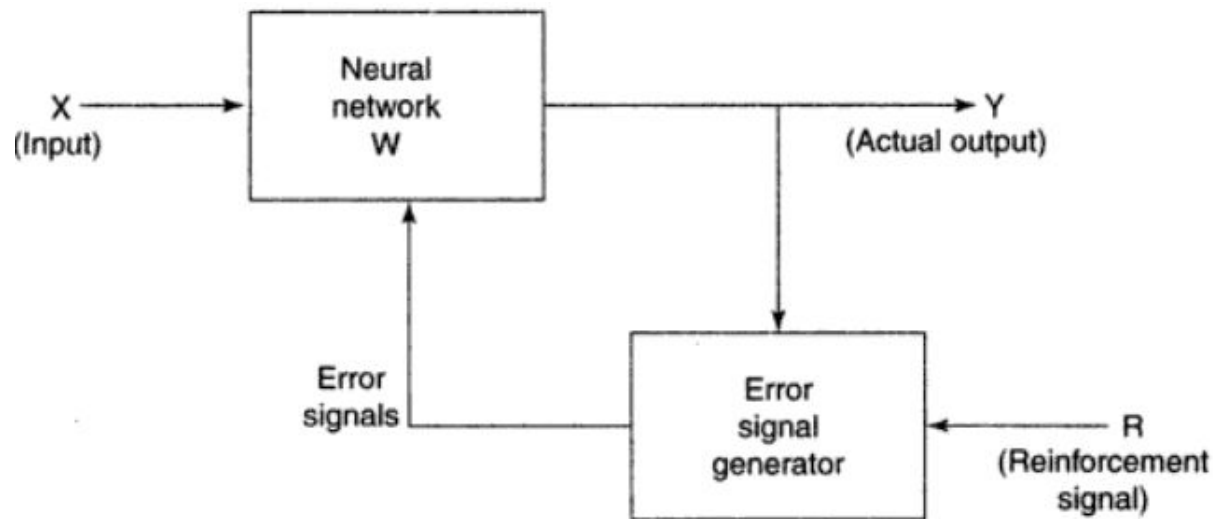
UNSUPERVISED LEARNING

- ✓ Learning is performed without the help of a teacher.
- ✓ Example: tadpole – learn to swim by itself.
- ✓ In ANN, during training process, network receives input patterns and organize it to form clusters.
- ✓ From the Fig. it is observed that no feedback is applied from environment to inform what output should be or whether they are correct.
- ✓ The network itself discover patterns, regularities, features/ categories from the input data and relations for the input data over the output.
- ✓ Exact clusters are formed by discovering similarities & dissimilarities so called as *self – organizing*.



REINFORCEMENT LEARNING

- ✓ Similar to supervised learning.
- ✓ Learning based on **critic information** is called *reinforcement learning* & the feedback sent is called *reinforcement signal*.
- ✓ The network receives some feedback from the environment.
- ✓ Feedback is only evaluative.
- ✓ The external reinforcement signals are processed in the critic signal generator, and the obtained critic signals are sent to the ANN for adjustment of weights properly to get critic feedback in future.



ACTIVATION FUNCTIONS

- ✓ To make work more efficient and for exact output, some force or activation is given.
- ✓ Like that, activation function is applied over the net input to calculate the output of an ANN.
- ✓ Information processing of processing element has two major parts: input and output.
- ✓ An integration function (f) is associated with input of processing element.
- ✓ The nonlinear activation function is used to ensure that a neuron's response is bounded
- ✓ When a signal is fed through a multilayer network with a linear activation functions the output obtained remains the same as that could be obtained using a single layer network
- ✓ *Due to this reason , nonlinear functions are widely used in multilayer network compared to linear function*

✓ Several activation functions are there.

1. Identity function:

it is a linear function which is defined as

$$f(x) = x \text{ for all } x$$

The output is same as the input.

2. Binary step function

it is defined as

$$f(x) = \begin{cases} 1 & \text{if } x \geq \theta \\ 0 & \text{if } x < \theta \end{cases}$$

where θ represents thresh hold value.

It is used in single layer nets to convert the net input to an output that is binary. (0 or 1)

3. *Bipolar step function:*

It is defined as

$$f(x) = \begin{cases} 1 & \text{if } x \geq \theta \\ -1 & \text{if } x < \theta \end{cases}$$

where θ represents threshold value

used in single layer nets to convert the net input to an output that is bipolar (+1 or -1).

4. *Sigmoid function*

used in Back propagation nets.

Two types:

a) binary sigmoid function

-logistic sigmoid function or unipolar sigmoid function.

-it is defined as

$$f(x) = \frac{1}{1 + e^{-\lambda x}}$$

where λ – steepness param

-The derivative of this function is

$$f'(x) = \lambda f(x)[1-f(x)].$$

The range of sigmoid function is 0 to 1.

b) Bipolar sigmoid function

$$f(x) = \frac{2}{1 + e^{-\lambda x}} - 1 = \frac{1 - e^{-\lambda x}}{1 + e^{-\lambda x}}$$

where λ - steepness parameter and the sigmoid range is between -1 and +1.

- The derivative of this function can be

$$f'(x) = \frac{\lambda}{2} f(x)[1-f(x)]$$

- It is closely related to hyperbolic tangent function, which is written as

$$h(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$

$$h(x) = \frac{1 - e^{-2x}}{1 + e^{-x}}$$

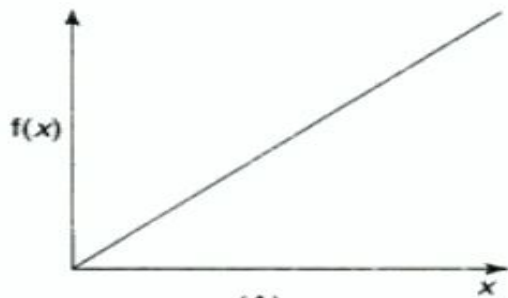
The derivative of the hyperbolic tangent function is

$$**h'(x) = [1+h(x)][1-h(x)]**$$

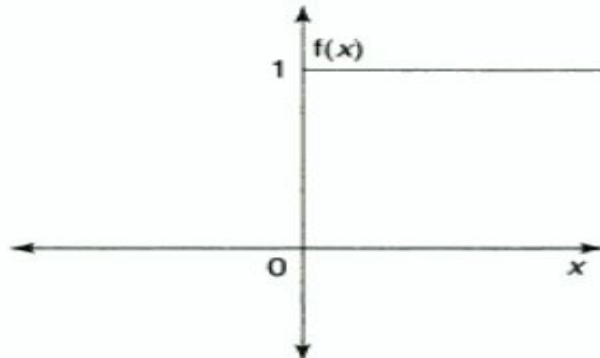
5. Ramp function

$$f(x) = \begin{cases} 1 & \text{if } x > 1 \\ x & \text{if } 0 \leq x \leq 1 \\ 0 & \text{if } x < 0 \end{cases}$$

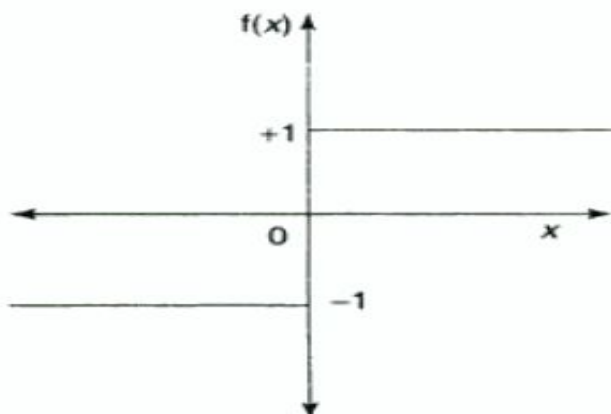
The graphical representation of all these function is given in the upcoming Figure



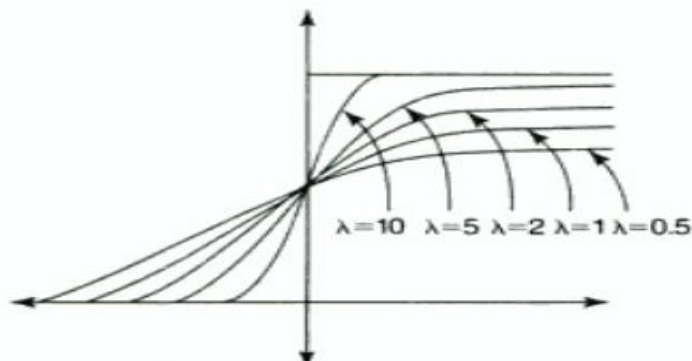
(A)



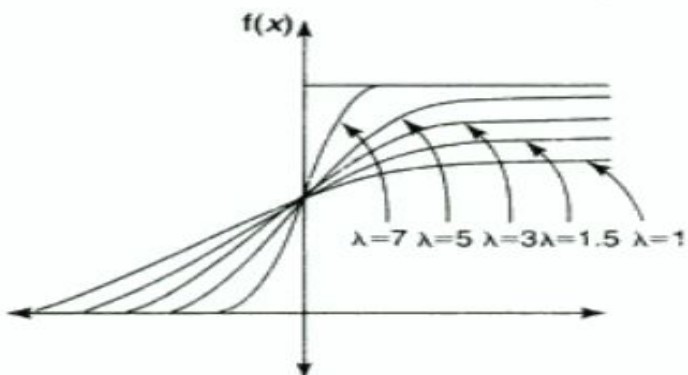
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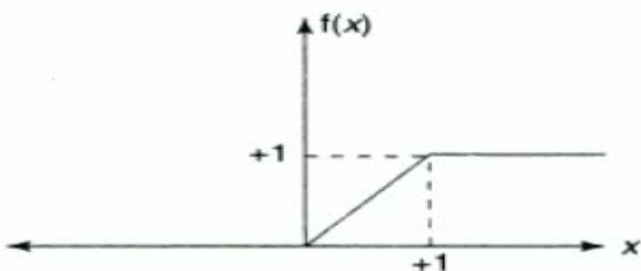
(C)



(D)



(E)



(F)

IMPORTANT TERMINOLOGIES

- **Weight**

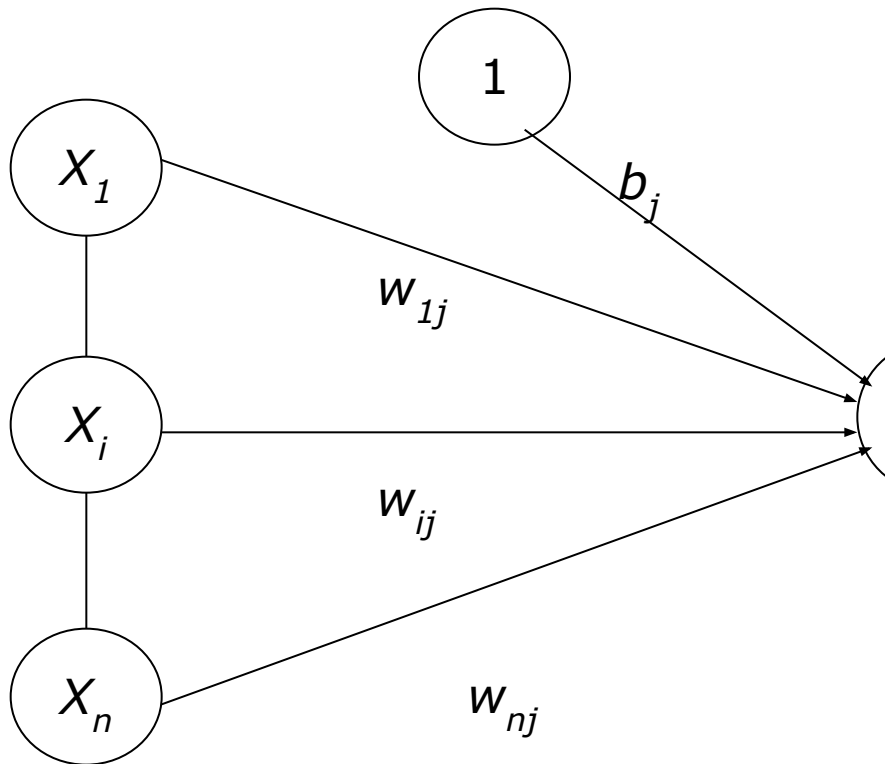
- The weight contain information about the input signal.
- It is used by the net to solve the problem.
- It is represented in terms of matrix & called as *connection matrix*.
- If weight matrix W contains all the elements of an ANN, then the set of all W matrices will determine the set of all possible information processing configuration.
- The ANN can be realized by finding an appropriate matrix W .
- Weight encode long-term memory (LTM) and the activation states of network encode short-term memory (STM) in a neural network.

Weight matrix

$$\mathbf{W} = \begin{bmatrix} \mathbf{w}_1^T \\ \mathbf{w}_2^T \\ \mathbf{w}_3^T \\ \vdots \\ \vdots \\ \vdots \\ \vdots \\ \vdots \\ \mathbf{w}_n^T \end{bmatrix} = \begin{bmatrix} \mathbf{w}_{11} & \mathbf{w}_{12} & \mathbf{w}_{13} & \cdots & \mathbf{w}_{1m} \\ \mathbf{w}_{21} & \mathbf{w}_{22} & \mathbf{w}_{23} & \cdots & \mathbf{w}_{2m} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \mathbf{w}_{n1} & \mathbf{w}_{n2} & \mathbf{w}_{n3} & \cdots & \mathbf{w}_{nm} \end{bmatrix}$$

Weights contd...

- w_{ij} is the weight from processing element "i" (source node) to processing element "j" (destination node)



$$y_{inj} = \sum_{i=0}^n x_i w_{ij}$$

$$= x_0 w_{0j} + x_1 w_{1j} + x_2 w_{2j} + \dots + x_n w_{nj}$$

$$y_j = w_{0j} + \sum_{i=1}^n x_i w_{ij}$$

$$y_{inj} = b_j + \sum_{i=1}^n x_i w_{ij}$$

Bias

- Bias has an impact in calculating net input.
- Bias is included by adding x_0 to the input vector x .
- The net output is calculated by

$$y_{in_j} = \sum_{i=0}^n x_i w_{ij}$$

$$y_{in_j} = b_j + \sum_{i=0}^n x_i w_{ij}$$

- The bias is of two types
 - Positive bias
 - » Increase the net input
 - Negative bias
 - » Decrease the net input

Threshold

- It is a set value based upon which the final output is calculated.
- Calculated net input and threshold is compared to get the network output.
- The activation function of threshold is defined as

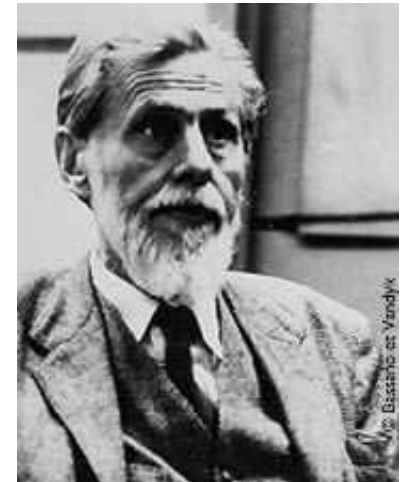
$$f(net) = \begin{cases} 1 & \text{if } net \geq \theta \\ -1 & \text{if } net < \theta \end{cases}$$

- where θ is the fixed threshold value

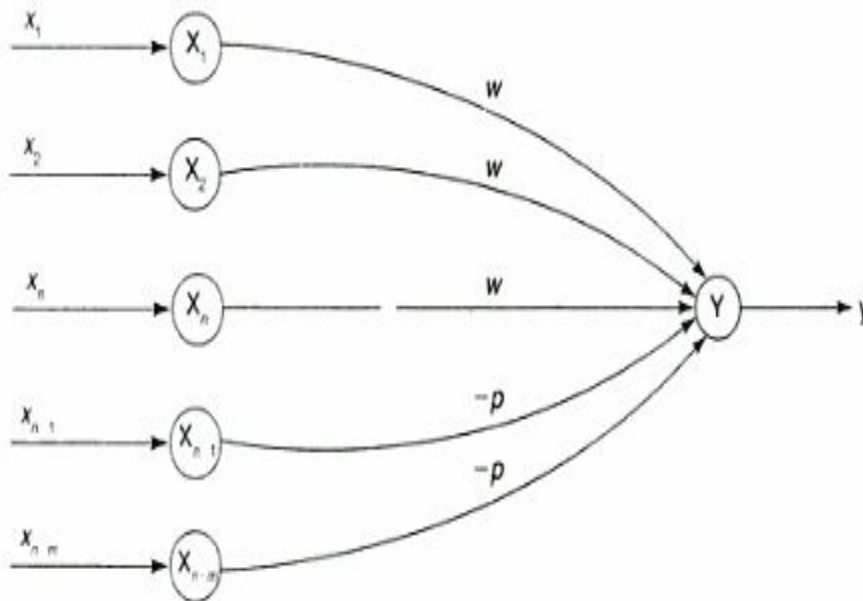
- **Learning rate**
 - Denoted by α .
 - Control the amount of weight adjustment at each step of training.
 - The learning rate range from 0 to 1.
 - Determine the rate of learning at each step
- **Momentum Factor**
 - Convergence is made faster if a momentum factor is added to the weight updation process.
 - Done in back propagation network.
- **Vigilance parameter**
 - Denoted by ρ .
 - Used in Adaptive Resonance Theory (ART) network.
 - Used to control the degree of similarity.
 - Ranges from 0.7 to 1 to perform useful work in controlling the number of clusters.

MCCULLOCH-PITTS NEURON

- Discovered in 1943.
- Usually called as *M-P neuron*.
- M-P neurons are connected by directed weighted paths.
- Activation of M-P neurons is binary (i.e) at any time step the neuron may fire or may not fire.
- Weights associated with communication links may be excitatory(wgts are positive)/inhibitory(wgts are negative).
- Threshold plays major role here. There is a fixed threshold for each neuron and if the net input to the neuron is greater than the threshold then the neuron fires.
- They are widely used in logic functions.



- A simple M-P neuron is shown in the figure.
- It is excitatory with weight ($w > 0$) / inhibitory with weight $-p$ ($p < 0$).
- In the Fig., inputs from x_1 to x_n possess excitatory weighted connection and x_{n+1} to x_{n+m} has inhibitory weighted interconnections.
- Since the firing of neuron is based on threshold, activation function is defined as



$$f(y_{in}) = \begin{cases} 1 & \text{if } y_{in} \geq \theta \\ 0 & \text{if } y_{in} < \theta \end{cases}$$

- For inhibition to be absolute, the threshold with the activation function should satisfy the following condition:

$$\theta > \sum w_i - p$$

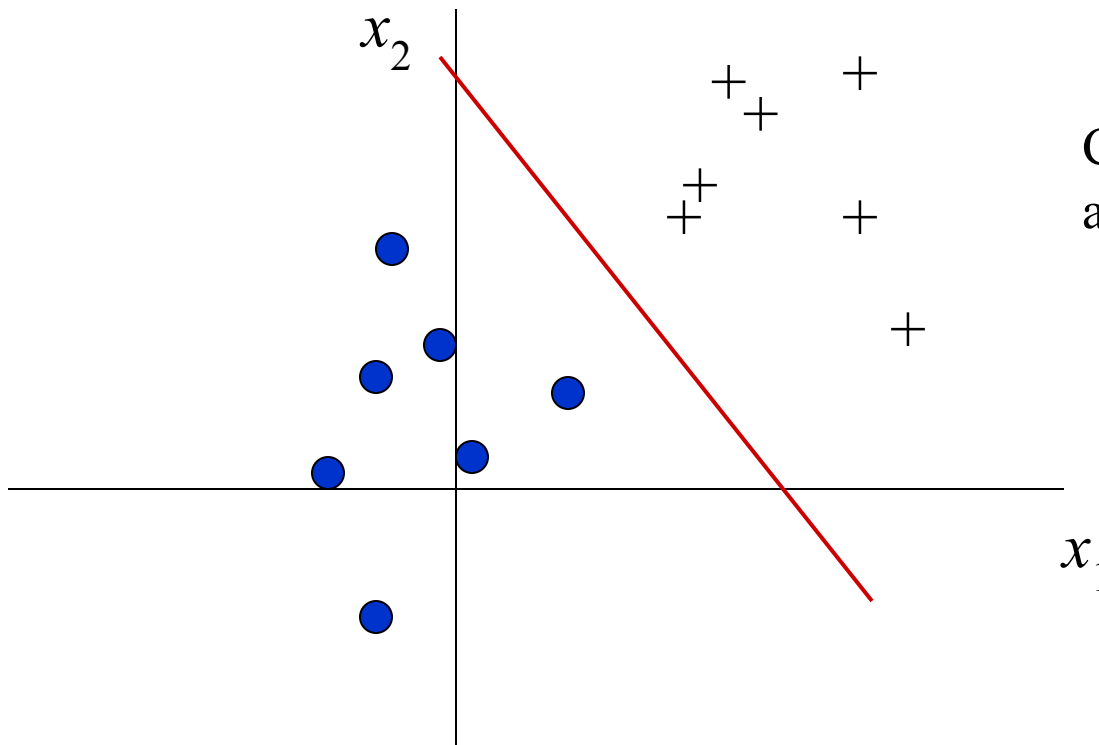
- Output will fire if it receives “ k ” or more excitatory inputs but no inhibitory inputs where

$$\sum w_i \geq \theta > (k-1) \sum w_i$$

- ✓ The M-P neuron has no particular training algorithm.
- ✓ An analysis is performed to determine the weights and the threshold.
- ✓ It is used as a building block where any function or phenomenon is modeled based on a logic function.

Linear Separability

Consider example with two inputs, x_1 , x_2 :



Can view trained network as defining a “**separation line**”.

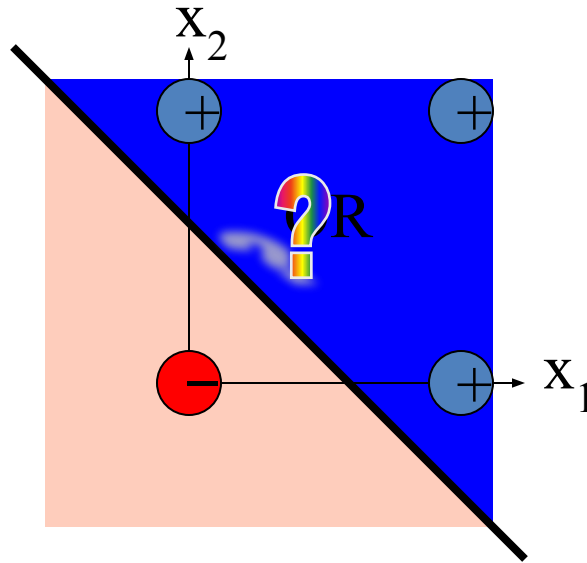
What is its equation?

$$-w_0 + w_1x_1 + w_2x_2 = 0$$

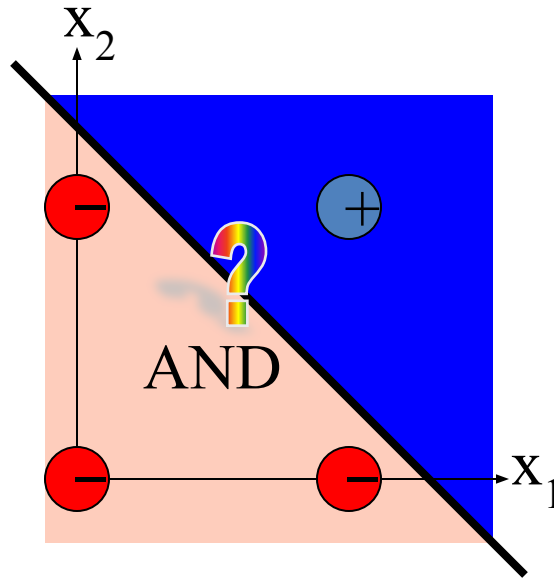
$$x_2 = -\frac{w_1}{w_2}x_1 + \frac{w_0}{w_2}$$

Perceptron used for classification

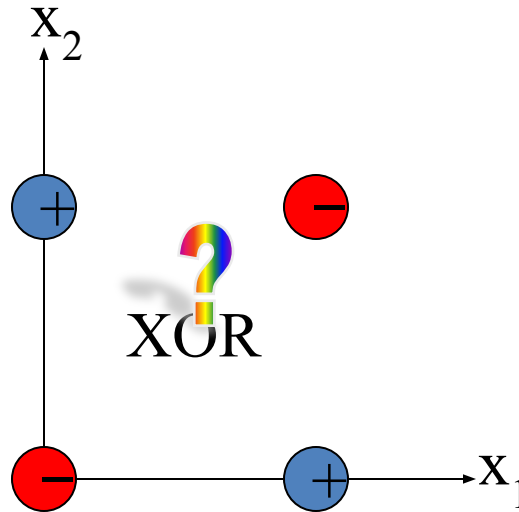
Linear Separability



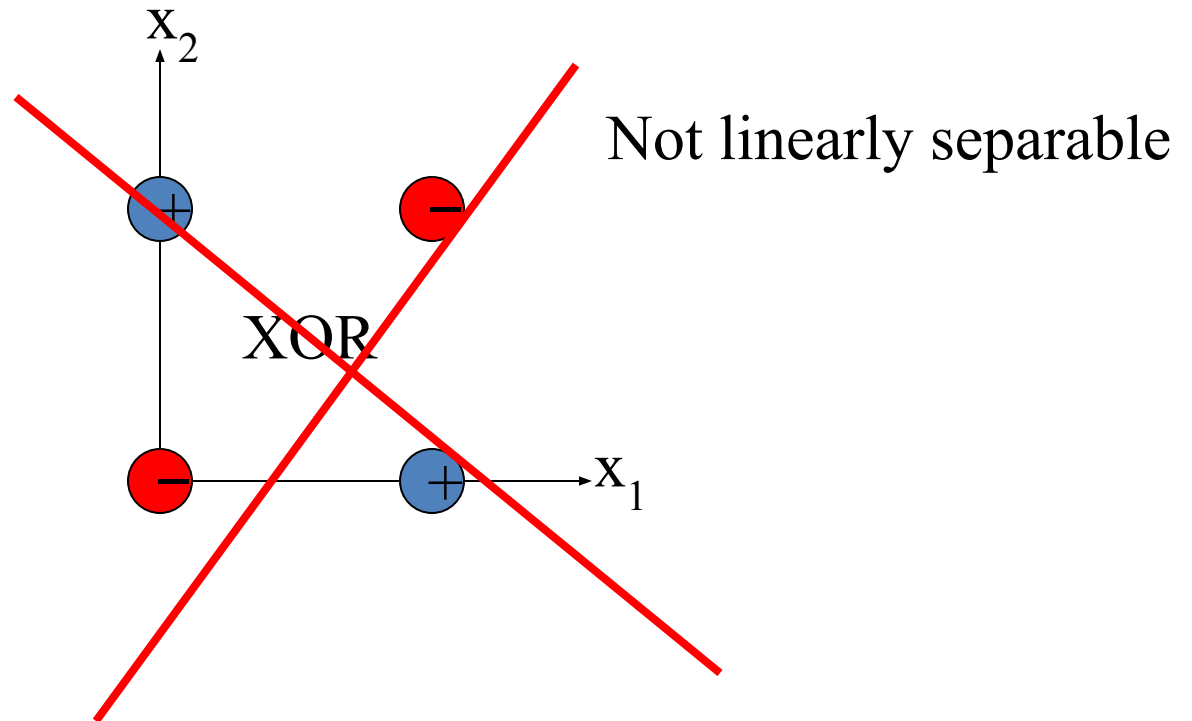
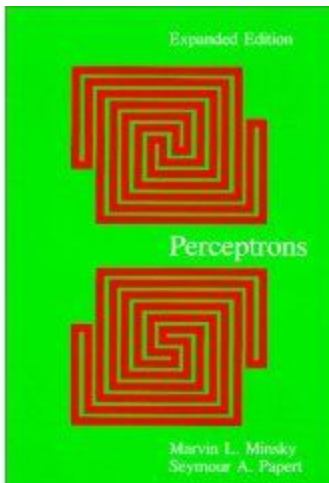
Linear Separability



Linear Separability



Linear Separability



Minsky & Papert (1969)

Bad News: Perceptrons can only represent linearly separable functions.

LINEAR SEPARABILITY

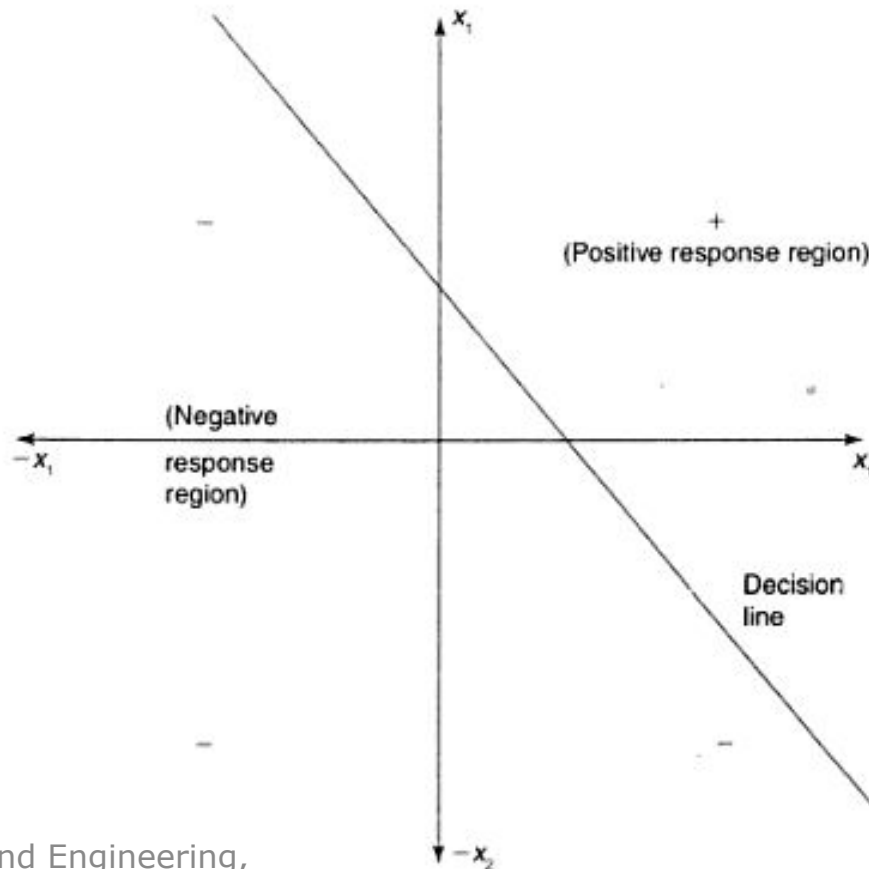
- ✓ It is a concept wherein the separation of the input space into regions is based on whether the network response is positive or negative.
- ✓ A decision line is drawn to separate positive or negative response.
- ✓ The decision line is also called as *decision-making line* or *decision-support line* or *linear-separable line*.
- ✓ The net input calculation to the output unit is given as

$$y_{in} = b_j + \sum_{i=1}^n x_i w_i$$

- ✓ The region which is called as *decision boundary* is determined by the relation

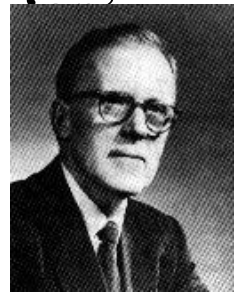
$$b + \sum_{i=1}^n x_i w_i = 0$$

- ✓ Consider a network having positive response in the first quadrant and negative response in all other quadrants with either binary or bipolar data.
- ✓ Decision line is drawn separating two regions as shown in Fig.
- ✓ Using bipolar data representation, missing data can be distinguished from mistaken data. Hence bipolar data is better than binary data.
- ✓ Missing values are represented by 0 and mistakes by reversing the input values from +1 to -1 or vice versa.

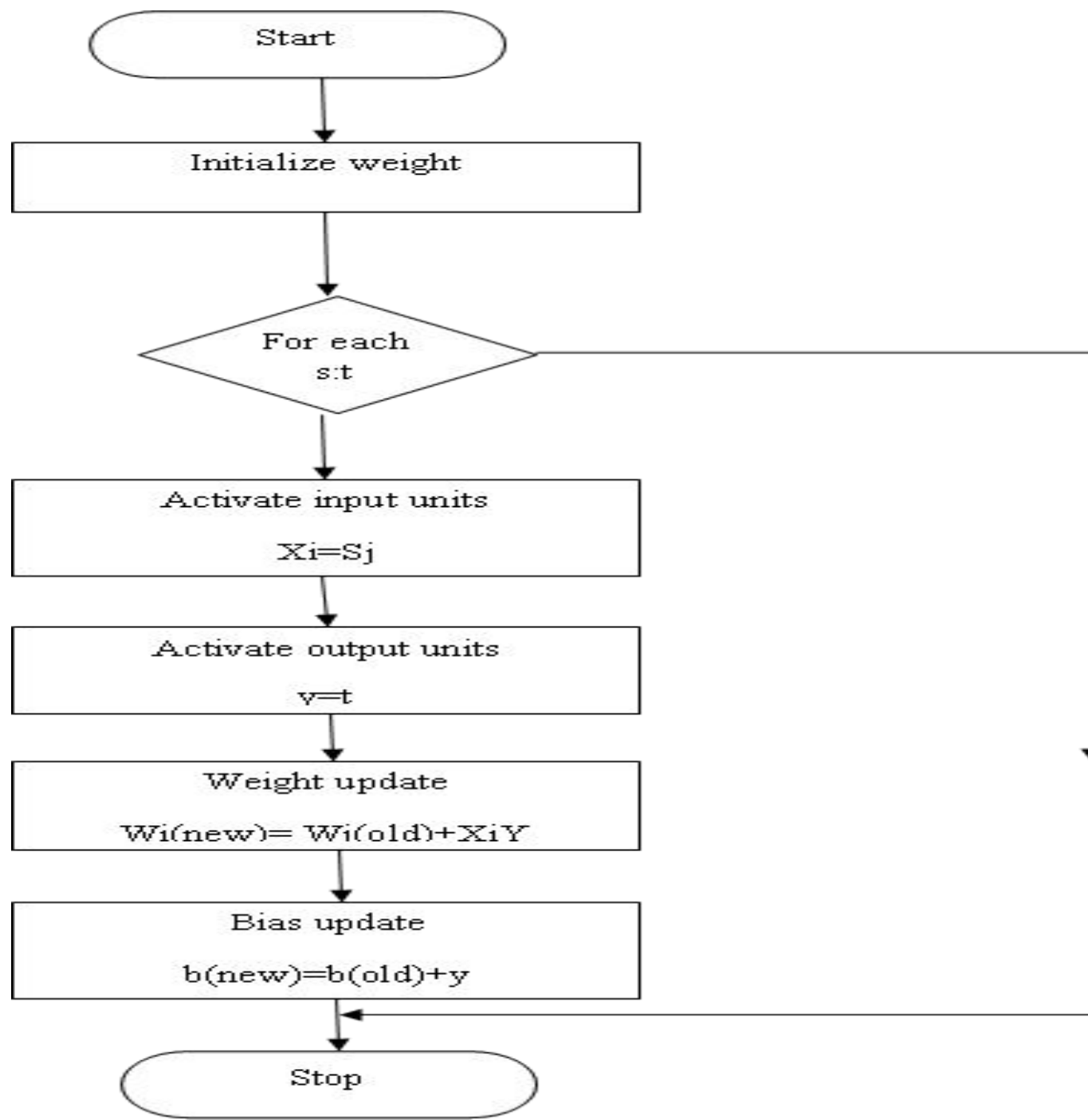


HEBB NETWORK

- Donald Hebb stated in 1949 that “ *In brain, the learning is performed by the change in the synaptic gap*”.
- When an axon of cell A is near enough to excite cell B, and repeatedly or permanently takes place in firing it, some growth process or metabolic change takes place in one or both the cells such that A's efficiency, as one of the cells firing B, is increased.
- According to Hebb rule, *the weight vector is found to increase proportionately to the product of the input and the learning signal*.
- In Hebb learning, two interconnected neurons are ‘ ‘ simultaneously.
- The weight update in Hebb rule is given by
 » $W_i(\text{new}) = w_i(\text{old}) + x_i y.$
- It is suited more for bipolar data.
- If binary data is used, the weight updation formula cannot distinguish two conditions namely
 - A training pair in which an input unit is “on” and the target value is “off”
 - A training pair in which both the input unit and the target value is “off”.



FLOWCHART OF TRAINING ALGORITHM



- Steps:
 - **0: First initialize the weights.**
 - **1: Steps 2-4 have to be performed for each input training vector and target output pair, s:t**
 - **2: Input activations are set. The activation function for input layer is identity function.**
 - $X_i = S_i$ for $i=1$ to n
 - **3: Output activations are set.**
 - **4: Weight adjustment and bias adjustments are performed.**
 - $W_i(\text{new}) = w_i(\text{old}) + x_i y$
 - $b(\text{new}) = b(\text{old}) + y$
- ✓ In step 4, the weight updating formula can be written in vector form as
 - ✓ $w(\text{new}) = w(\text{old}) + y.$
- ✓ Change in weight is expressed as
 - ✓ $\Delta w = xy$
 - ✓ Hence,
 - ✓ $w(\text{new}) = w(\text{old}) + \Delta w$
 - ✓ Hebb rule is used for pattern association, pattern categorization, pattern classification and over a range of other areas

- Construct a feed forward network with five input nodes, three hidden nodes and four output nodes

- Construct a feed forward network with five input nodes, three hidden nodes and four output nodes that has lateral inhibition structure in the output layer

- Construct a recurrent network with 4 input nodes, three hidden nodes and 2 output nodes that has a feedback link from hidden layer to input layer

Using Hebb rule, Find the weight required to perform the following classification of the given input patterns shown below

The pattern is shown as 3X3 matrix form in the squares. The “+” symbol represents the value 1 and empty square indicate -1.

Consider “I” belongs to the member of the class so the target value is 1 and “O” does not belongs to the member of the class so the target value is -1

+	+	+
	+	
+	+	+

“I”

+	+	+
+		+
+	+	+

“O”

Using Hebb rule, Find the weight required to perform the following classification of the given input patterns shown below

The pattern is shown as 3X3 matrix form in the squares. The “+” symbol represents the value 1 and dot indicate -1.

Consider “L” belongs to the member of the class so the target value is 1 and “U” does not belongs to the member of the class so the target value is -1

+	.	.
+	.	.
+	+	+

“L”

+	.	+
+	.	+
+	+	+

“U”

Using Hebb rule, Find the weight required to perform the following classification of the given input patterns shown below

The pattern is shown as 3X3 matrix form in the squares. The “+” symbol represents the value 1 and dot indicate -1.

Consider “A” belongs to the member of the class so the target value is 1 and “E” does not belongs to the member of the class so the target value is -1

+	+	+
+	.	+
+	+	+
+	.	+
+	.	+

“A”

+	+	+
+	.	.
+	+	+
+	.	.
+	+	+

“E”

Using Hebb rule find weights required to perform following classifications. The vectors $(1 \ -1 \ 1 \ -1)$ and $(1 \ 1 \ 1 \ -1)$ belongs to class (target value +1); vectors $(-1 \ -1 \ 1 \ 1)$ and $(1 \ 1 \ -1 \ -1)$ do not belongs to class(target value -1). Also using each of training x vectors as input, test the response of the net

Distinguish between pattern “X” and pattern “O”

#				#
	#		#	
		#		
	#		#	
#				#

	#	#	#	
#				#
#				#
#				#
	#	#	#	